Experiment 7: Pole-Zero and Stability Analysis of ContinuousTime Systems

Input coding

```
clc;
clear;
close all;

num = [1 1];
den = [1 4 5];

% Find poles and zeros
poles = roots(den);
zeros = roots(num);

disp('Poles:');
disp(poles);

disp('Zeros:');
disp(zeros);

% Pole-Zero Plot
figure;
plot(real(poles),imag(poles),'x','MarkerSize',10,'LineWidth',2);
hold on;
plot(real(zeros),imag(zeros),'o','MarkerSize',10,'LineWidth',2);
grid on;

xlabel('Real Axis');
ylabel('Imaginary Axis');
title('Pole-Zero Plot');
legend('Poles','Zeros');

% Stability Check
if all(real(poles)<0)
    disp('System is Stable');
else
    disp('System is Unstable');
end
```

Output

